Organoid Brain-on-a-Chip Systems: Resource Map, Bioelectronic Interface Strategy and First-Demo Roadmap

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Abstract

Brain-on-a-chip research is converging with human brain organoids, microfluidics, multielectrode arrays (MEAs), soft bioelectronics and biological computing. This survey maps high-value resources and turns them into a buildable roadmap. The key engineering thesis is simple: biology connects to electronics through a stable extracellular interface, not through direct digital access to cells. A feasible first demo should therefore avoid wet-lab complexity and prove the loop in software first: spike-like data, preprocessing, burst/network features, a decoder or reservoir readout, a stimulation policy, and a safety gate. Wet biology can then be added through a remote platform or commercial MEA before attempting custom 3D organoid chips.

Keywords: brain organoid; brain-on-chip; organoid intelligence; wetware computing; microelectrode array; microfluidics.

1 Scope

This document treats "brain on a chip with organoids" as four overlapping systems: brain organoids maintained in engineered devices, brain organoids interfaced to MEAs or soft electrodes, closed-loop biological computing platforms, and software/data resources for analysing organoid activity. It is a high-confidence research map, not a claim that every marginal paper in the broader organoid literature has been enumerated.

2 Landscape

Three technical bottlenecks dominate. First, organoids need vascular, metabolic, immune and extracellular-matrix support to mature reproducibly. Microfluidics helps by adding flow, oxygenation, gradients and repeatable dosing. Second, planar MEAs undersample spherical tissues. New devices therefore wrap, embed, suspend or penetrate organoids using shells, meshes, liquid metals and multilayer electrodes. Third, organoid computing requires closed-loop stimulation, stable readout, shared data formats and defensible interpretation of learning-like behaviour.

3 Core Literature

4 Repositories

5 Projects and Platforms

6 How Biology Connects to Electronics

The connection is extracellular and analog. Neurons and glia remain alive in culture medium; electrodes only sense voltage changes in the nearby ionic solution and inject controlled current or voltage pulses back into that solution. A workable system therefore has five layers: viable tissue, a mechanical/chemical interface, electrodes, low-noise electronics, and software that converts spikes or field potentials into decisions.

7 First Demo

The first demo should be software-only and biologically honest: no organoid culture, no custom chip fabrication, and no claim of learning. The aim is to prove that the repository can run the complete control loop on realistic spike data. Use MEA-NAP-style channel spike times [10], FinalSpark-style remote notebooks when access exists [31, 28], or the Cortical Labs SDK/API examples for local development of closed-loop applications [33, 34]. Closed-loop stimulation reviews emphasize that the controller must be faster than the biological process and that the chosen feature must be controllable [21].

Demo acceptance criteria. A run should produce a single report with: input dataset hash or simulator seed, channel map, detected bursts, selected stimulation events, safety-limit decisions, latency estimate, and plots. Passing means the loop is reproducible and explainable, not that the organoid has learned.

8 Roadmap to Wet Biology

The path becomes manageable if each phase has one new uncertainty. Phase 1 tests algorithms. Phase 2 tests real data. Phase 3 tests vendor APIs. Phase 4 tests living neural tissue without custom fabrication. Phase 5 tests custom organoid geometry and 3D interfaces. This sequence follows the direction of emerging bioelectronics reviews: start with MEA-compatible measurements, then improve geometric coupling and chronic stability [20].

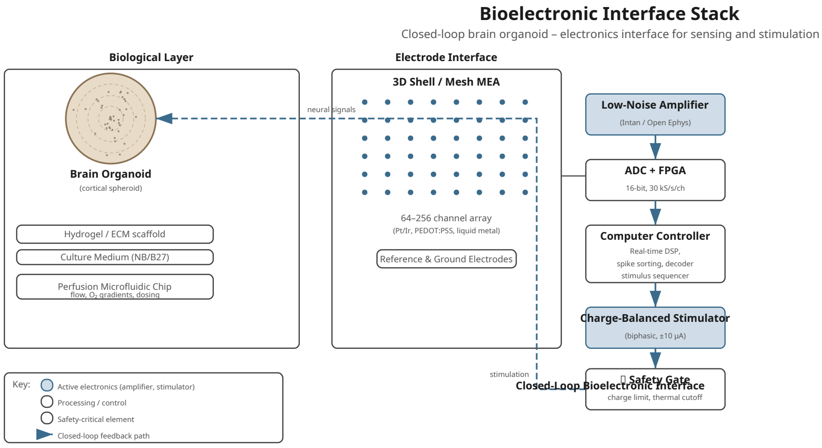

Figure 1: Feasible bioelectronic stack. The cell culture is never "plugged into" a computer directly. The electronics observe extracellular voltage, and the controller writes back through safe stimulation pulses.

9 Research Gaps

The next useful project should not start by claiming general intelligence. It should build a reproducible loop: standardized organoid culture, 3D MEA geometry, stimulation protocol, common data format, analysis notebook, and ethics log. The most publishable near-term experiments are comparative: planar MEA versus shell/mesh/3D MEA; static culture versus perfused culture; open-loop stimulation versus adaptive closed-loop stimulation; single-organoid versus organoid-array ensembles.

10 Recommended Repository Direction

For this repository, the most valuable structure is:

- `references/papers/`: PDFs and citation metadata.
- `references/repos/`: vendored snapshots or submodules for code references.
- `publications/`: this survey and later manuscripts.
- `experiments/`: future notebooks for MEA signal analysis, reservoir computing, and closed-loop simulations.
- `data/`: only small public example datasets; keep large electrophysiology data external with manifest files.

11 Companion Repository

OrganoidVision is the companion repository for retinal-organoid sensing and bio-hybrid vision systems [42]. OrganoidIntelligence should remain the broad roadmap for brain-on-chip intelligence, while OrganoidVision develops the application branch for organoid sensors, optical stimulation, event-based vision and sensor-stack blueprints.

12 Conclusion

Brain-organoid-on-chip systems are best understood as engineered interfaces around living neural tissue. The field's practical center is disease modelling, toxicology and electrophysiology. Organoid intelligence adds a computing agenda, but its credibility depends on careful controls, repeatable closed-loop protocols, and open analysis. The strongest resource path for this repository is to combine curated papers, reproducible MEA tools, and explicit project tracking for FinalSpark, Cortical Labs, NSF BEGIN OI, and open organoid data efforts.

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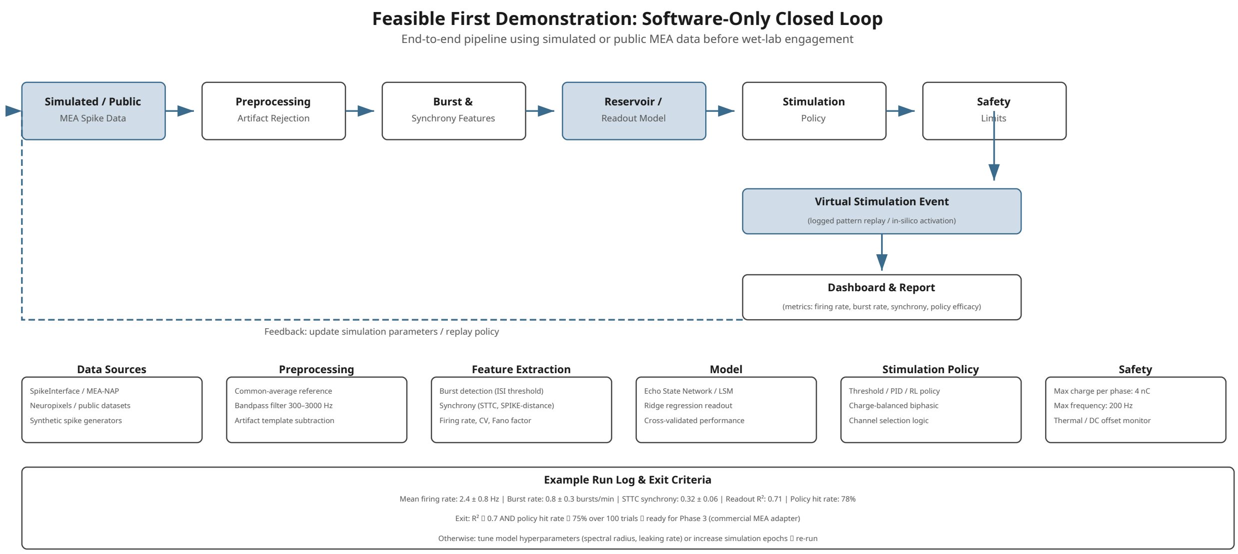

Figure 2: Minimum viable demo. It proves the digital side of the wetware loop before any living tissue is introduced.

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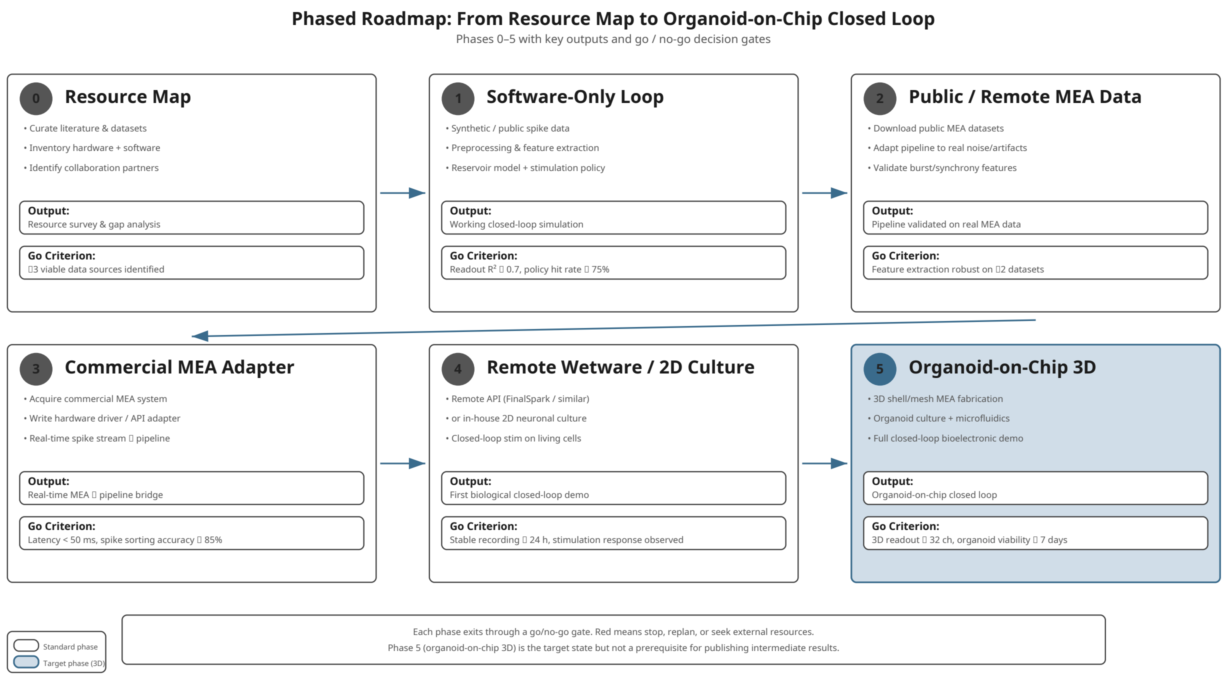

Figure 3: Phased roadmap. Each phase keeps the previous phase's software and replaces only one layer: simulator, data adapter, hardware adapter, then living tissue.

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Table 1: Curated papers and reviews that anchor the organoid brain-on-chip topic.

Theme	Resource	Why it matters
Foundation	Lancaster et al., cerebral organoids, 2013 [1]	Establishes self-organized cerebral organoids as human brain development models.
On-chip folding	Karzbrun et al., human brain organoids on a chip, 2018 [2]	Uses chip-constrained organoids to study mechanics of cortical folding.
Toxicology chip	Wang et al., prenatal nicotine exposure, 2018 [3]	Early hiPSC brain organoid-on-chip disease/toxicology platform.
Microfluidic assembly	Ao et al., prenatal cannabis exposure, 2020 [4]	One-stop microfluidic brain organoid assembly and exposure model.
Maturation	Cho et al., brain ECM plus microfluidic flow, 2021 [5]	Shows brain ECM and dynamic microfluidics improve survival, maturation and electrophysiology.
Neuroinflammation	Ao et al., immune-competent/tubular organoids, 2021 [6]	Adds microfluidic culture strategies for inflammatory modelling.
Preclinical review	Castiglione et al., 2022 [7]	Focused review of human brain organoids-on-chip for preclinical use.
Disease review	Zhang et al., 2025 [8]	Recent review of history, challenges and neural disease modelling trends.
Integration review	Zhao et al., 2024 [9]	Broad Nature Reviews map of organoid–organ-on-chip integration.
MEA pipeline	MEA-NAP paper, 2024 [10]	Practical network analysis pipeline for 2D cultures and 3D brain organoids.
HD-MEA readout	Functional imaging using high-density MEAs, 2022 [11]	Demonstrates single-unit and network measurements in human cerebral organoids.
3D MMF	Park et al., 3D multifunctional neural interfaces, 2021 [12]	Flexible 3D devices for cortical spheroids and engineered assembloids.
Cyborg organoids	Li et al., 2019; Le Floch et al., 2022 [13, 14]	Mesh nanoelectronics integrated through organogenesis for chronic tissue-wide recordings.
Shell MEA	Huang et al., 2022 [15]	Self-folding shell electrodes encapsulate brain organoids for 3D surface readout.
Liquid-metal interface	Wu et al., 2024 [16]	Stretchable 128-channel liquid-metal mesh interface for hippocampal organoids.
Reshapable MEA	Magnetically reshapable liquid-metal 3D MEAs, 2025 [17]	Adjustable soft 3D electrodes for intra-organoid electrophysiology.
Depth MEA	Kim et al., multilayered MEA, 2026 [18]	Depth-resolved multilayer recordings across cerebral organoid vertical structure.
Full-coverage interface	Liu et al., shape-conformal porous frameworks, 2026 [19]	Nearly full surface coverage, high-channel-count electrophysiology and programmed stimulation for neural organoids.
Organoid intelligence	Smirnova et al., OI, 2023 [22]	Defines organoid intelligence as a field combining brain organoids and machine interfaces.
Ethics roadmap	Baltimore Declaration and OI workshop, 2023 [23, 24]	Establishes governance and community framing for OI.
Brainoware	Cai et al., 2023 [25]	Brain organoid reservoir computing for speech recognition and nonlinear prediction.
DishBrain	Kagan et al., 2022 [26]	Closed-loop cultured-neuron system embodied in a Pong-like environment.
Sample efficiency	Khajehnejad et al., 2024 [27]	Compares biological neural cultures with deep RL agents in a simulated game.
Remote platform	FinalSpark Neuroplatform paper, 2024 [28]	Remote API/Jupyter access to living neuronal/organoid electrophysiology.
Computing overview	Talavera and Ulmann, 2025 [29]	Broad overview of brain organoid computing and bibliography.
Tactile encoding	Liu et al., 2025/2026 [30]	Braille-like tactile classification using stimulated forebrain organoids.

Table 2: Code and data repositories worth tracking or vendoring as references.

Repository	Type	Use
`github.com/SAND-Lab/MEA-NAP`	MEA analysis	Network topology, burst, connectivity and batch analysis for MEA recordings.
`github.com/Cortical-Labs/cl-api-doc`	Platform docs	Early CL1 API notebooks for closed-loop neuron experiments.
`github.com/Cortical-Labs/cl-sdk`	SDK/simulator	Python simulator for Cortical Labs API workflows.
`github.com/Cortical-Labs/SpikeStream`	Visualization	Real-time spike-stream visualization code.
`github.com/TBC-TJU/MetaBCI`	BCI software	Open-source BCI algorithms and stimulation/brainflow tooling already vendored locally.
`github.com/quadbio/primate_cerebral_organoids`	Reproducibility	Code for primate cerebral organoid single-cell analyses.
`github.com/quadbio/organoid_regulomes`	Reproducibility	Cell-fate regulome analysis in human cerebral organoids.
`github.com/utokyoIkeuchilab/neural-organoid-activity-MEA-analysis`	MEA analysis	Activity analysis for connected cerebral organoids.
`github.com/danielathome19/pyorganoid`	Simulation	Python package for organoid intelligence and organoid learning simulation.
`github.com/McD-NMI/OrganoidMEA`	Simulation/analysis	Electrophysiology simulation and analysis for OrganoidMEA.
`github.com/Unlimited-Research-Cooperative/human-cortical-organoid-intelligence-system`	Concept/demo	Human cortical organoid intelligence system experiments and notes.
`github.com/aimarket/awesome-autonomous-drone-racing`	Robotics reference	Locally vendored robotics/RL reference; useful only for downstream closed-loop embodied tasks.

Table 3: Active projects, companies, platforms and programs.

Name	Role	Notes
FinalSpark Neuroplatform	Remote wetware computing	Provides 24/7 remote access, neural stimulation/recording, Python API and large recorded datasets.
Cortical Labs CL1 / Cortical Cloud	Commercial biological computer	Code-deployable neuronal computing platform with public API docs and SDK.
NSF BEGIN OI	Funding program	US EFRI program for Biocomputing through EnGINeering Organoid Intelligence.
OpenOrganoid	Open data/community	Open platform for organoid research data and AI-assisted organoid analysis.
Johns Hopkins OI community	Research community	Source of OI workshop, Baltimore Declaration and organoid intelligence roadmap.
Axion BioSystems Maestro / Organoid MEA	Instrumentation	Commercial MEA tools and protocols for neural organoid functional readout.
Multi Channel Systems / Mesh MEA	Instrumentation	Mesh MEA product line for recording from within 3D organoids.
NETRI NeuroFluidics MEA	Microfluidic MEA	Standardized organs-on-chip combining microfluidics with Axion-compatible MEA readout.
MIMETAS OrganoPlate	Organ-on-chip platform	Scalable perfused microfluidic culture platform; broader organoid-on-chip relevance.
Emulate Brain-Chip	Brain-on-chip platform	Microphysiological brain-barrier/neural platform, relevant to non-organoid brain-chip baselines.

Table 4: Build decisions that reduce the gap between biology and electronics.

Layer	Practical choice	Why it makes the project feasible
Biology	Start with public data, a simulator, or a remote platform before local organoids	Removes cell-culture risk while validating the computational loop.
Recording	Use planar MEA data first; move to shell, mesh or 3D MEA only after analysis works	Planar MEA is simpler and has mature tools; 3D electrodes solve coverage later.
Stimulation	Use biphasic, charge-balanced pulses and strict software limits	Reduces electrode damage, electrochemistry and biological stress risk.
Signal path	Separate recording, stimulation, artifact blanking and feature extraction	Prevents the first demo from confusing stimulation artifact with biological response.
Control	Begin with rule-based burst feedback, then add reservoir/readout models	Gives an interpretable baseline before machine-learning claims.
Safety	Log every pulse, amplitude, duration, channel, tissue state and result	Makes the experiment auditable and easier to reproduce.

Table 5: Concrete implementation plan for the first two weeks.

Step	Deliverable	Implementation detail
Day 1–2	Repository skeleton	Add `experiments/first-demo/`, `data/manifests/`, and a README describing simulator-first scope.
Day 3–4	Spike simulator	Generate Poisson plus bursting spike trains across 32–64 channels with fixed random seeds.
Day 5–6	Feature extractor	Bin spikes, detect bursts, compute firing rate, inter-burst interval and channel synchrony.
Day 7–8	Closed-loop policy	Trigger a virtual stimulation event when burst rate crosses a threshold or when synchrony drops.
Day 9–10	Safety manager	Reject events that violate refractory interval, maximum pulses per minute or channel blacklist.
Day 11–12	Dashboard/report	Save PNG/PDF plots and a machine-readable JSON run log.
Day 13–14	Optional real-data adapter	Add importer for MEA-NAP-style outputs, FinalSpark notebook exports or Cortical Labs SDK streams.